



Implementation Guide for UICS Updates 2017

December 2017

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1 About this Document

1.1 Scope

This Implementation Guide incorporates updates on the *UnionPay Integrated Circuit Card Specifications* for 2017. The updates involve both Basic Specifications and Auxiliary Specifications of the UICS, particularly in support of mobile payment product upgrading and international market expansion. This Implementation Guide includes the following chapters:

Business Requirements — The updates arising from new business requirements are introduced to the Specifications to support payment product innovation and business development on both domestic and international markets.

Technical Enhancements — Technical updates are described to facilitate Issuers and Acquirers to better implement UnionPay IC card products.

Specification Enhancements — In addition to the specification enhancements in 2017, all the functional changes introduced in 2015 and 2016 are also incorporated.

1.2 References

UnionPay Integrated Circuit Card Specifications – Basic Specifications

UnionPay Integrated Circuit Card Specifications – Auxiliary Specifications

Implementation Guide for UICS Updates 2015

Implementation Guide for UICS Updates 2016

UICS Updates of ODA for Online Authorization

Mobile Payment Specifications

1.3 Summary of Updates

The changes and their implementation requirements, effective dates, and affected institutions and regions are outlined in the following table.

Change	Implementation Requirement	Effective Date	Institution	Region
Requirements for Supporting CDCVM and Two-Tap Features	Mandatory, applied to new contactless acceptance terminals	Jan. 1, 2018	Acquirers	Outside of Mainland China
	Mandatory, applied to existing contactless acceptance terminals	Jun. 30, 2020		
Updated Solution of U.S. Common Debit AID	Conditional, if UnionPay debit cards and certain pre-paid cards are issued	Jan. 1, 2018	Issuers	U.S.
Updated Card Personalization Templates	Mandatory	Jan. 1, 2018	Issuers	Global
Specifications Structural Change – Removal of the Contact Terminal Specification	Mandatory, applied to contact terminals	Upon Publication	Acquirers	Global
Updated Definition of Product Identification Information	Mandatory	Upon Publication	Issuers	Global
Introduced Description of Remote Updating Cardholder Personal Data	Optional	Upon Publication	Issuers	Global
Incorporation of Updates in 2015 and 2016	Conditional ¹	Conditional ¹	Conditional ¹	Conditional ¹
Specification Enhancements in 2017	Conditional ²	Upon Publication	Conditional ²	Global

Note:

Conditional¹ – For details of the updates in 2015 and 2016, please refer to the Implementation Guides in 2015, 2016 and the “UICS Updates of ODA for Online Authorization” accordingly.

Conditional² – For detailed impact of the enhancements in 2017, please refer to the Section 4.2 of this document.

2 Business Requirements

2.1 Requirements for Supporting CDCVM and Two-Tap Features

2.1.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)			Outside of Mainland China
	IC card (contactless)			
Acquirers	Terminal (contact)			
	Terminal (contactless)	Mandatory	√	

2.1.2 Background

The CDCVM (Consumer Device Cardholder Verification Method), introduced in the *Implementation Guide of UICS Updates 2015*, verifies the user identity of the mobile device and confirms the Cardholder legitimacy using biometric factors and/or knowledge factors, etc. The two-tap feature was introduced in the *Implementation Guide of UICS Updates 2016*. When CDCVM is performed on mobile payment devices, the two-tap feature indicates that two taps of the user's mobile device on the terminal are required to complete the transaction.

In view of the fast-growing volume of transactions initiated from mobile devices, UnionPay International (UPI) focuses on bringing faster speed and greater convenience to Cardholders. To improve the performance of the mobile NFC payment product and maintain a consistent and secure Cardholder experience outside of Mainland China, UPI updates the UICS requirements for the contactless acceptance terminals on supporting the CDCVM and two-tap features.

2.1.3 Function Description

CDCVM – Byte 3 Bit 7 in the Terminal Transaction Qualifiers (tag '9F66') is enabled to indicate whether the CDCVM is supported by the terminal. If the CDCVM is a mutually supported CVM between the terminal and the mobile payment device, mobile NFC payment transactions shall continue only when the consumer completes verification on the mobile device.

Two-Tap – When the CDCVM is required for the transaction but has not been performed on the mobile payment device, the device shall return Status Word '6986' in a GPO response to the terminal. Then the terminal will prompt the user to follow the instructions on the payment device, power off the contactless interface immediately, and wait for 1,000ms to 1,500ms to power on the contactless interface again.

2.1.4 Requirements for Institutions

Requirements for Contactless Terminals Manufacturers:

- Effective January 1, 2018, all new contactless acceptance terminals must support CDCVM and two-tap features.

Requirements for Acquirers:

- Effective January 1, 2018, Acquirers shall purchase and deploy new contactless acceptance terminals that comply with the UICS 2017.
- Effective June 30, 2020, Acquirers shall complete the upgrading of existing contactless acceptance terminals to support the CDCVM and two-tap features.

2.1.5 Test and Certification Requirements

Effective January 1, 2018, new contactless acceptance terminals must support the CDCVM and two-tap features and pass the related functional tests.

2.2 Updated Solution of U.S. Common Debit AID

2.2.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)	Conditional, if UnionPay debit cards and certain pre-paid cards are issued	√	U.S.
	IC card (contactless)			
Acquirers	Terminal (contact)			
	Terminal (contactless)			

2.2.2 Background

The U.S. Common Debit AID was introduced in UICS in 2015 to provide the U.S. Merchants and Acquirers with a more flexible routing choice in compliance with the Durbin Amendment of the United States. In order to implement the issuing of debit cards and certain prepaid cards in the U.S., an enhanced solution of U.S. Common Debit AID has been developed.

2.2.3 Changes to the Specifications

■ Basic Specifications – Part II Debit/Credit Application Card Specification

Enabled Issuer Identification Number (tag '42') for U.S. Common Debit AID

Name (format; tag; length; source)	Requirement	Description	Value
Issuer Identification Number (IIN) F: n6 T: 42 L: 3 Source: Card	O	The number that identifies the major industry and the card Issuer and that forms the first part of the Primary Account Number (PAN). It is referred to as the BIN in this document. In template 'BF0C' or '73'. It is returned to the terminal in SELECT or READ RECORD response.	

■ Auxiliary Specifications – Part I Personalization Template Based on Debit/Credit Application

UPI has added a new personalization template for contact debit cards issued in the U.S. The new Template 10 is designed based on Template 1 with the following differences:

- a) Introduced Issuer Identification Number (tag '42') in Template 10;
- b) Introduced Application Identifier (tag '4F') and Application Label (tag '50') of the U.S. Common Debit AID in Template 10;
- c) Revised CVM List (tag '8E') synchronously with the updating of Template 1. Refer to Section 2.3 for details.

Details of the three differences are presented below. For a full description of Template 10, please refer to Section 4.10 in the *UICS Auxiliary Specifications Part I*.

1. Card Data Object in Template 10

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Application Identifier (AID)	b	'4F'	5-16	Initialized	UnionPay debit card AID: A0 00 00 03 33 01 01 01 UnionPay debit cards and some prepaid cards issued in the U.S. shall support the U.S. Common Debit AID: A0 00 00 03 33 01 01 08 <i>Note: If the U.S. Common Debit AID is supported, the two AIDs above shall point to the same debit application.</i>				
Application Label	ans	'50'	1-16	Filled in by Issuers according to the actual situation.	Application name when the terminal displays an optional application list to the consumer.		☐		

Data Name	Data Format on Card	Tag	Data Length (bytes)	Value (Hexadecimal)	Description	Template Default Setting	Issuer General Data	Card or Cardholder Specific Data	Data Stored in File Record
Cardholder Verification Method (CVM) List	b	'8E'	14	0000 0000 0000 0000 4203 1E03 1F00	Lists all Cardholder verification methods supported by the card application in order of priority. <i>Note: An application can have multiple CVM lists, for example: one is for domestic transactions and the other is for cross-border transactions.</i>	☐			☐
Issuer Identification Number (IIN)	n	'42'	3		The number that identifies the major industry and the card Issuer and that forms the first part of the Primary Account Number (PAN). It is referred to as BIN in this document. In template 'BF0C' or '73'. It is returned to the terminal in SELECT or READ RECORD response.				

2. Application Identifier and Application Label

Application Identifier (AID)		Application Label
Registered Application Provider Identifier (RID)	Proprietary Identifier Extension (PIX)	Card Type
A0 00 00 03 33	01 01 01	UnionPay Debit
	01 01 08	U.S. Common Debit <i>Note: The Application Label of the U.S. Common Debit AID must not contain "UnionPay".</i>

3. Cardholder Verification Method (CVM) List

'0000 0000 0000 0000 4203 1E03 1F00' hexadecimal

CVM Code – first eight bytes are 0s	CVM	Processing Order	Condition	Action if CVM Fails
0100 0010 0000 0011	Online PIN	1	If supported by the terminal	Apply succeeding CVM
0001 1110 0000 0011	Signature	2	If supported by the terminal	Cardholder verification fails
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

2.2.4 Requirements for Institutions

Institution	Requirement
Issuers	Issuers in the U.S. shall comply with the latest requirement if they issue UnionPay debit cards and/or certain pre-paid cards.
Acquirers	N/A

2.2.5 Test and Certification Requirements

Effective January 1, 2018, the personalization test of the newly issued U.S. UnionPay debit cards and some pre-paid cards shall follow the requirement of the UICS 2017.

2.3 Updated Card Personalization Templates

2.3.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)	Mandatory	√	Global
	IC card (contactless)			
Acquirers	Terminal (contact)			
	Terminal (contactless)			

2.3.2 Background

Card Additional Processes (tag '9F68') is one of the critical card parameters for contactless transaction process. To better assist the Issuer in personalizing the contactless card accurately in compliance with the latest UICS requirements and improve the Cardholder experience in both domestic and international contactless transactions, UnionPay has updated the personalization Templates 11- 16 to clarify the settings of Card Additional Processes for both PIN-based and signature-based contactless cards.

For mobile device personalization, apart from the UICS Personalization Template, Issuers shall also refer to the *UnionPay Mobile Payment Specification* or contact UPI mobile product team for specific requirements.

To regulate the CVM setting of debit cards, Template 1 is updated. In addition, a new U.S. only debit card template (Template 10) is introduced for convenience of the U.S. Issuers (refer to Section 2.2).

2.3.3 Changes to the Specifications

■ Auxiliary Specifications – Part I Personalization Template Based on Debit/Credit Application

- Further detailed the settings of Card Additional Processes (tag '9F68') in Templates 11-16. Listed in the table below are items that require strict compliance. For detailed values and definitions in the tag '9F68', please refer to the Personalization Template.

➤ CVM is online PIN

Byte	Bit	Value	Definition
1	2	0	The card does not prefer to choose contact IC card online transaction first
2	7	0	Non-currency-matching transaction allowed
3	8	1	Currency-matching transaction supporting online PIN
	7	1	Non-currency-matching transaction supporting online PIN

Byte	Bit	Value	Definition
	6	0	For non-currency-matching transactions, CVM not required
	5	1	Signature supported
4	6	1	ODA supported for online authorization

➤ CVM is signature

Byte	Bit	Value	Definition
1	2	0	The card does not prefer to choose contact IC card online transaction first
2	7	0	Non-currency-matching transaction allowed
3	8	0	Currency-matching transaction not supporting online PIN
	7	0	Non-currency-matching transaction not supporting online PIN
	6	0	For non-currency-matching transactions, CVM not required
	5	1	Signature supported
4	6	1	ODA supported for online authorization

- Revised the CVM List in Template 1 so that the signature is supported by UnionPay debit cards, and thus bypass PIN is allowed.

Cardholder Verification Method (CVM) List

'0000 0000 0000 0000 4203 1E03 1F00' hexadecimal

CVM Code – first eight bytes are 0	CVM	Processing Order	Condition	Action if CVM Fails
0100 0010 0000 0011	Online PIN	1	If supported by the terminal	Apply succeeding CVM
0001 1110 0000 0011	Signature	2	If supported by the terminal	Cardholder verification fails
0001 1111 0000 0000	CVM not required	3	Always	Will not fail

- Added Template 10 so that UnionPay debit cards and pre-paid cards issued in the U.S. comply with local regulations. Please refer to 2.2.3 for details.

2.3.4 Requirements for Institutions

Institution	Requirement
Issuers	Issuers shall personalize the card according to the Personalization Template of UICS 2017.
Acquirers	N/A

2.3.5 Test and Certification Requirements

Effective January 1, 2018, the personalization test of the newly issued cards shall follow the requirements of the UICS 2017

2.4 Specifications Structural Change – Removal of the Contact Terminal Specification

2.4.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)			Global
	IC card (contactless)			
Acquirers	Terminal (contact)	Mandatory	√	
	Terminal (contactless)			

2.4.2 Background

UICS Basic Specifications Part III Debit/Credit Application Terminal Specification is fully compatible with *China Financial IC Card Specification Part 6* and *EMV IC Card Specification for Payment Systems Book 3*. To simplify the reference for the contact terminal kernel and its processes, the contact terminal specification is removed from the UICS. The Acquirers as well as contact terminal manufacturers inside Mainland China shall refer to *China Financial IC Card Specification Part 6*, while the Acquirers and contact terminal manufacturers outside of Mainland China shall refer to *EMV IC Card Specifications for Payment Systems*¹ for requirements on contact terminals of debit/credit application.

The UICS continues:

- Keeping UnionPay members and vendors informed of the changes of UnionPay chip cards (contact and contactless) and mobile NFC payment products.
- Specifying UnionPay's minimum technical requirements for chip-based debit/credit programs.
- Providing implementation options for members and vendors to meet market needs.
- Providing UnionPay's definitions of card brand-specific features reserved by EMV specifications.

¹ The latest *EMV IC Card Specifications for Payment Systems* and its related bulletins can be downloaded from <https://www.emvco.com/emv-technologies/contact/>, or acquired by contacting the UPI technical specifications team at ts@unionpayintl.com.

2.4.3 Changes to the Specifications

■ Basic Specifications – Part III Debit/Credit Application Terminal Specification

Removed Part III of the UICS.

■ Basic Specifications – Part II Debit/Credit Application Card Specification

Appendix A incorporated terminal data elements from Part III and added the “Source” tag in the table of data elements to provide a global view for audience to easily identify the data source.

In addition, the description of Issuer Script Processing was moved from Part III to Part II as Appendix K to maintain the readability of the UICS.

2.4.4 Requirements for Institutions

Institutions	Requirements
Issuers	N/A
Acquirers	For the requirements on contact terminal of debit/credit applications, the Acquirers inside Mainland China shall refer to <i>China Financial IC Card Specification Part 6</i> , and the Acquirers outside of Mainland China shall refer to <i>EMV IC Card Specification Book 3</i> .

2.4.5 Test and Certification Requirements

The test and certification process remains unchanged. The contact acceptance terminals inside Mainland China shall pass the PBOC Level 2 tests; while the contact acceptance terminals outside of Mainland China shall pass the EMV Level 2 tests.

3 Technical Enhancements

3.1 Updated Definition of Product Identification Information

3.1.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)	Mandatory	√	Global
	IC card (contactless)			
Acquirers	Terminal (contact)			
	Terminal (contactless)			

3.1.2 Background

Product Identification Information (tag '9F63') indicates the form factor of the payment instruments used in the transaction, the type of the contactless interface over which the transaction is conducted, and the additional information used to specify the details of the mobile NFC payment.

Product Identification Information is defined in the UICS and some of its bytes reserved for mobile NFC payment are specified in the *UnionPay Mobile Payment Specifications*. To help members and vendors better understand the characteristics of mobile NFC payment, the reserved definition of Product Identification Information (tag '9F63') is synchronized in the UICS, and its requirement is changed from 'O' to 'C (if it is a mobile device card)'.

3.1.3 Changes to the Specifications

■ Basic Specifications – Part II Debit/Credit Application Card Specification

Appendix A.1: Specified Product Identification Information (tag '9F63') according to the definition in the *UnionPay Mobile Payment Specifications*.

Product Identification Information F: b 128 T: 9F63 L: 16 Source: Card	C If it is a mobile device card.	Used to identify the Issuer and card product type, and sent to the Issuer in the online messages.	Byte 1-8: Bank Identification Number Byte 9: Card Product Identification Bit 8:1=Citizen card Bit 7:1=Veteran card Bit 6:1=Bonus card Bit 5:1=Traffic card Bit 4:1=Social security card Bit 3:1=Student card Bit 2:1= Air card Bit 1:1=Public affair toll card Byte 10: Product Form Indicator
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			Bits 3-1: Version number of product indicator; and version No. 00 is used in this Specifications. The value is 000b, and the other values are reserved. Bits 8-4: 00001= Standard chip card 00010=Innovative card based on the standard card form (e.g., display card, Bluetooth card) 00011=Innovative card based on the non-card form (e.g., wristband, watch, key fob, etc.) 00100=Mobile device card based on HCE 00101=Mini card (physical chip card, which is smaller than a standard ID-1 card) 10000=Mobile device card based on SE Others: RFU Byte 11: Product Subdivision Identification When Byte 10 is 0x10, the definition of Byte 11 is as follows: 0x01=Display card (standard size) 0x02=Bluetooth card (standard size) Others: RFU When Byte 10 is 0x18, the definition of Byte 11 is as follows: Bits 8-5 indicate the communication interface; Bits 4-1 indicate the product form. Detailed value is as follows: 0001 0000=Bluetooth non-standard innovative card (e.g., pendant, mobile phone pendant, key ring pendant, etc.) 0001 0001=Smart watch wearable device 0001 0010=Smart bracelet wearable device 0010 0000=Audio interface payment plug-in <i>Note: The Apple Pay product is included in this definition.</i> Others: Reserved When Byte 10 is 0x20, the definition of Byte 11 is as follows: Bits 8-5: RFU Bit 4: 1=SE-involved Bit 3: 1=Token-involved Bit 2: 1=TEE-involved Bit 1: 1=Cloud-involved When Byte 10 is 0x80 and it indicates the mobile device card based on SE, the definition of 11th byte is as follows:
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			0001 0000=SWP-SD card 0001 0001=Dual-interface SD card 0001 0010=Pure remote SD card 0010 0000=SWP-SIM card 0010 0001=Dual-interface SIM card 0010 0010= Pure remote SIM card 0011 0000=Whole mobile phone 1011 0000=iPhone 1011 0001=Apple Watch 1011 0010=iPad Others: Reserved Byte 12: Carrier Issuer Definition Bits 8-1: 0010 0000=China Mobile 0010 0001=China Telecom 0010 0010=China Unicom 0100 0000=Apple 0100 0001=Huawei 0100 0010=Xiaomi 0100 0011=Samsung 0100 0100=ZTE 0100 0101=Lenovo 0100 0110=Codoon 0100 0111=G&D 0100 1000=Oppo 0100 1001=Ginoee 0100 1010=LeEco 0100 1011=Meitu 0100 1100=Meizu 0100 1101=Smartisan 0100 1110=Vivo 0100 1111=Qiku(360) 0101 0000=Coolpad 0110 0000=Lakala 0110 0001=Watchdata 0110 0010=Goldpac 0110 0011=Mobvoi 0110 0100=Hengbao 0110 0101=Shenzhen Ce-link 0110 0110=Datang Microelectronics
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			0110 0111=Shenzhen Huiyun Technology 0110 1000=Eastcompeace 0110 1001=Chengxin Data 0110 1010=Garmin(6A) 0110 1011=Peacewear 0110 1100=HiCling 0110 1101=PayTend 0110 1110=Swatch 0110 1111=Okline 0111 0000=Tianyu Others: Reserved Byte 13: Product Device Feature Bit 8: 1=Passcode capable (different from PIN used in financial transactions, the passcode on payment device is used to protect financial transactions) Bit 7: 1=Signature panel Bit 6: 1=Hologram Bit 5: 1=CVN2 Bit 4: 1=Dual authentication (the payment device is capable of exchanging identifying message between the Issuer and Cardholder, of which the method is defined by the Issuer, e.g., SMS confirmation) Bits 3-1: RFU (000) Byte 14: Reserved by the bankcard association Bytes 15-16: Reserved by the Issuer
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3.1.4 Requirements for Institutions

Institution	Requirement
Issuers	Issuers shall accurately set this tag on the mobile device according to either the <i>UnionPay Mobile Payment Specification</i> or the UICS.
Acquirers	Acquirers shall send this tag online if the terminal reads it out.

3.1.5 Test and Certification Requirements

The current test plan is valid and no additional test is required.

3.2 Introduced Description of Remote Updating Cardholder Personal Data

3.2.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)	Optional	√	Global
	IC card (contactless)			
Acquirers	Terminal (contact)			
	Terminal (contactless)			

3.2.2 Background

To protect the cardholder sensitive data, the *Implementation Guide of UICS updates 2015* has introduced the new requirement that “The Issuer should consider protecting the cardholder’s sensitive information, it is recommended not to personalize authentic user information, such as full cardholder name, into the card”. Besides above requirement for new issued cards, UICS introduced an appendix to illustrate remote update solution for existing card to further protect the cardholder sensitive information.

3.2.3 Changes to the Specifications

■ Basic Specifications – Part II Debit/Credit Application Card Specification

Appendix L (Descriptive Appendix) Description of Remote Updating Cardholder personal data

3.2.4 Requirements for Institutions

Institution	Requirement
Issuers	It’s optional for Issuer to implement. Issuer should arrange the card data of issued card to identify that the cardholder personal data present on which cards and the record number used to store the personal data. Issuer should choose one of the method to update and desensitize the personal data. After sending the UPDATE RECORD command, issuer should check the script execution result
Acquirers	Acquirer host systems should be able to send the issuer script command to the terminal and send back the execution result back to UnionPay

3.2.5 Test and Certification Requirements

N/A

4 Specification Enhancements

4.1 Incorporation of Updates in 2015 and 2016

4.1.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)	Depend on each update.	Depend on each update.	Depend on each update.
	IC card (contactless)			
Acquirers	Terminal (contact)			
	Terminal (contactless)			

Note: For implementation requirements, impacts and affected regions of each update, please refer to the Implementation Guides in 2015 and 2016, and the "UICS Updates of ODA for Online Authorization" accordingly.

4.1.2 Background

To present a full and comprehensive version of the UICS, UPI has incorporated the functional changes introduced in 2015 and 2016 into the UICS 2017. The following section lists these updates in each part of the UICS as well as the references for such information as their implementation requirements and impacts.

4.1.3 Change to the Specifications

■ Basic Specifications –Part I Debit/Credit Application Overview

No.	Item	Section	Description	Reference
1.	Transaction Flow Adjustment	4.1.11, 4.1.12	Switched the order of Section 4.1.11 and Section 4.1.12, and updated the transaction process order in Figure 1.	Implementation Guide of UICS Updates 2015
		4.2.11.6	Updated the transaction process order in Figure 16: switched the order of "Terminal processes Issuer Script" and "Terminal completes transaction".	

■ Basic Specifications – Part II Debit/Credit Application Card Specification

No.	Item	Section	Description	Reference
1.	Common AID	3.1.1	Added a description about Common AID in the application selection.	Implementation Guide of UICS Updates 2015
2.	Sensitive Information Protection	16.2	Added a description about the access restriction of transaction log records.	Implementation Guide of UICS Updates 2015
		17.1.7	Updated the description about the load log.	
		A.1 A.2.9	Revised the requirement and description of Cardholder Name (tag '5F20'), Cardholder Name Extended (tag '9F0B') and Cardholder ID Number (tag '9F61').	

No.	Item	Section	Description	Reference
3.	PAR	A.1 A.2.9	Added the definition of PAR (tag '9F24') in Table A.1 and Table A.2.	<i>Implementation Guide of UICS Updates 2016</i>
4.	Quick Chip	Appendix I	Added Appendix I Quick Chip Transaction Process to describe the Quick Chip technical solution.	<i>Implementation Guide of UICS Updates 2016</i>
5.	Credit-mode Low-value Payment	Appendix J	Added Appendix J Credit-mode Low-value Payment to describe technical requirements on contactless credit-mode low-value payment.	<i>Implementation Guide of UICS Updates 2016</i>
6.	Application Label	4.1	Revised the description of Application Label (tag '50') in Table 4 Application Selection - Card Data.	<i>Implementation Guide of UICS Updates 2016</i>
		A.1	Revised the description of Application Preferred Name (tag '9F12'): "Application Preferred Name should be consistent with Application Label."	

■ Basic Specifications – Part IV Debit/Credit Application Security Specification

No.	Item	Section	Description	Reference
1.	ODA for Online Authorization	3.3.5.1, 3.3.5.2	Added '95' as a valid value of Signed Data Format in Table 13 and Table 15, and supplemented related requirements on the generation method and source data of dynamic signature to support ODA for online authorization.	<i>UICS Updates of ODA for Online Authorization</i>

■ Basic Specifications – Part V Contactless Integrated Circuit Card Payment Specification

No.	Item	Section	Description	Reference
1.	CDCVM	7.1	Enabled Byte 3 Bit 7 in Terminal Transaction Qualifiers (tag '9F66') to indicate whether the terminal supports CDCVM.	<i>Implementation Guide of UICS Updates 2015</i>
		8.4.3.3	Added a description to clarify the Cardholder verification process.	
		Appendix A	Enabled Byte 2 Bit 8 in Card Transaction Qualifiers (tag '9F6C') to indicate whether CDCVM is performed.	
2.	qUICS Online Authorization Supports ODA	7.1	Enabled Byte 1 Bit 1 in Terminal Transaction Qualifiers (tag '9F66') to indicate if the terminal supports ODA in qUICS online authorization.	<i>Implementation Guide of UICS Updates 2015</i>
		8.4.3	Enabled Byte 4 Bit 6 in Card Additional Processes (tag '9F68') to indicate if the card supports ODA in qUICS online authorization.	

No.	Item	Section	Description	Reference
		8.4.2.2	Added a new table to list GPO response data elements in qUICS ODA for online authorization.	
		8.4.2.4	Revised the description of qUICS online transaction processes.	
		4, 8, 9, 10	Revised the requirements of reading application data and performing offline data authentication	
		C.1	Revised the requirements of dynamic data signature generation and verification, and added signed data format '95'.	
3.	Sensitive Information Protection	Appendix A	Added sensitive information protection requirements in Cardholder Name (tag '5F20'), Cardholder Name Extension (tag '9F0B') and Cardholder ID number (tag '9F61')	<i>Implementation Guide of UICS Updates 2015</i>
4.	Two-Tap Feature for CDCVM	8.2, 8.3.2	Added descriptions of the scenario and terminal GPO response processing logic when the card returns '6986', and updated Figure 5 accordingly.	<i>Implementation Guide of UICS Updates 2016</i>
5.	PAR	Appendix A	Added the definition of PAR (tag '9F24') in Table A.1.	<i>Implementation Guide of UICS Updates 2016</i>
6.	Specifications Change	4.3.2, B	Deleted the content related to dCVN.	<i>Implementation Guide of UICS Updates 2016</i>

■ Auxiliary Specifications – Part I Personalization Template Based On Debit/Credit Application

No.	Item	Section	Description	Reference
1.	PAR	4.1.1, 4.2.1, 4.3.1, 4.4.1, 4.5.1, 4.6.1, 4.7.1, 4.9.1	Added the definition of PAR (tag '9F24') in card data elements in Templates 1-7, 9.	<i>Implementation Guide of UICS Updates 2016</i>
2.	Application Label	4.1.10, 4.2.10, 4.3.10, 4.4.10, 4.5.10, 4.6.10, 4.7.10, 4.9.10	Clarified the Application Label (tag '50') for UnionPay ICCs and other mobile payment products according to the credit/debit attribute of a specific product in Templates 1-7, 9.	<i>Implementation Guide of UICS Updates 2016</i>

No.	Item	Section	Description	Reference
3.	Support CDCVM	5.1.3.5, 5.2.2.5, 5.3.2.5, 5.4.2.5, 5.5.2.5, 5.6.2.5, 5.7.1.5	Enabled Byte 2 Bit 8 of Card Transaction Qualifiers (tag '9F6C') to indicate whether CDCVM is performed in Templates 11-17.	<i>Implementation Guide of UICS Updates 2016</i>
4.	Support ODA for Online Authorization	5.2.2.4, 5.3.2.4, 5.4.2.4, 5.5.2.4, 5.6.2.4, 5.7.1.4	Enabled Byte 4 Bit 6 of Card Additional Processes (tag '9F68') to indicate whether ODA for online authorization is supported in qUICS process; Deleted Byte 1 Bit 6 "Does not support low-value payment or CTTA check", and set this bit to RFU in Templates 12-17.	<i>Implementation Guide of UICS Updates 2016</i>
		5.2.2.7, 5.3.2.7, 5.4.2.7, 5.5.2.7, 5.6.2.7, 5.7.1.7	Enabled Byte 1 Bit 7 of Card Enhanced Function Indicator (tag 'DF61') to indicate whether ODA for online authorization is supported in qUICS process in Templates 12-17.	
5.	Contactless Credit-Mode Low-Value Payment	5.2.2.7, 5.3.2.7, 5.4.2.7, 5.5.2.7, 5.6.2.7, 5.7.1.7	Enabled Byte 1 Bit 5 of Card Enhanced Function Indicator (tag 'DF61') for contactless credit-mode low-value payment to indicate whether the card supports credit-mode low-value payment in Templates 12-17.	<i>Implementation Guide of UICS Updates 2016</i>
		4.1.13, 4.2.14, 4.3.14, 4.4.14, 4.5.14, 4.6.14, 4.7.14, 4.9.13	Enabled Byte 2 Bit 2 of Application Default Action (tag '9F52') to indicate whether the tag '9F79'/'DF79' will be reset automatically in GAC processing in Templates 1-7, 9.	
	Security Upgrading	5.1.2.5 5.2.1.5, 5.3.1.5, 5.4.1.5, 5.5.1.5, 5.6.1.5	Adjusted IAC-online and IAC-default in low-value payment based debit/credit templates. Set Byte 1 Bit 3 of IAC-online and IAC-default to 1 to ensure the transaction is transmitted for online processing in case of CDA failure in Templates 11-16.	<i>Implementation Guide of UICS Updates 2016</i>
6.	Contents Removal of "Low-Value Payment or CTTA Check"	5.1.3.4 5.2.2.4, 5.3.2.4, 5.4.2.4, 5.5.2.4, 5.6.2.4, 5.7.1.4	Deleted Byte 1 Bit 6 of Card Additional Processes (tag '9F68'), and set this bit to RFU in Templates 11-17.	<i>Implementation Guide of UICS Updates 2016</i>
7.	CVM Clarification in Template 17	5.7.1.4	Revised the setting of Card Additional Processes (tag '9F68') in Templates 11-17 for PIN-based cards and signature-based cards.	<i>Implementation Guide of UICS Updates 2016</i>

■ Auxiliary Specifications – Part II Debit/Credit Application - Member Implementation Guide for Issuer

No.	Item	Section	Description	Reference
1.	Specifications Change	7.2.1.1.1	Deleted “credit/debit application 0101” in Table 2 UPI Proprietary Application Identifier Extension (PIXs)	<i>Implementation Guide of UICS Updates 2016</i>

■ Auxiliary Specifications – Part III Debit/Credit Application - Member Implementation Guide for Acquirer

No.	Item	Section	Description	Reference
1.	Merchant Information Accuracy Requirements	12.2.2	Added a description to specify the Merchant information accuracy requirement.	<i>Implementation Guide of UICS Updates 2015</i>

■ Auxiliary Specifications – Part V Debit/Credit Application Personalization Guide

No.	Item	Section	Description	Reference
1.	Common AID	7.5	Added Section 7.5 to describe the Common AID personalization requirements.	<i>Implementation Guide of UICS Updates 2015</i>

4.2 Specification Enhancements in 2017

4.2.1 Impact Overview

Institution	Card/Terminal	Requirement	Impact	Region
Issuers	IC card (contact)	Depend on each update	Depend on each update	Global
	IC card (contactless)			
Acquirers	Terminal (contact)			
	Terminal (contactless)			

4.2.2 Background

Since the full version of the UICS was published in 2014, UPI has kept collecting feedback from both domestic and international industry participants. In the UICS 2017, UPI has removed the unused terms and added necessary explanations to make the Specifications more understandable. The majority of these enhancements are textual optimization, and thus have no impact on UPI Members, as indicated by ‘N/A’ in the ‘Impact’ column in the table below. However, necessary updates are required on Members’ systems in the case of non-compliance.

4.2.3 Changes to the Specifications

■ Basic Specifications – Part I Debit/Credit Application Overview

Change	Reason	Impact	Section
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Change	Reason	Impact	Section
Revised the description of ADF file to clarify the indication of the DF name of ADF file.	Enhance the Specifications.	N/A	3.1.1
Revised the description about "CVM not required".	Enhance the Specifications.	N/A	4.2.6.5.3.4
Revised the "Application Selection" flow chart: when the application is blocked, the card returns FCI and SW1 SW2 = '6283' in SELECT response.	Align the flow chart content with the context.	N/A	4.2.1.7
Deleted the note in Table 41 about terminal action after transaction completion.	Remove the content which is out of the scope of the UICS.	N/A	4.2.11.5

■ Basic Specifications – Part II Debit/Credit Application Card Specification

Change	Reason	Impact	Section
Revised the description and the flow chart about application selection: when the PSE/application is blocked, the card returns FCI and SW1 SW2 = '6283' in SELECT response.	Align the flow chart content with the context.	N/A	4.4.1 4.6
Added a description about Issuer authentication to further clarify that the card performs Issuer authentication when it receives the EXTERNAL AUTHENTICATE command no matter whether AIP indicates the command is supported or not.	Enhance the Specifications.	N/A	13.4.3
Revised the access requirement of Application Default Action (tag '9F52'), added it to the PUT DATA command so that it can be updated, and added it to the GET DATA (special device) command so that it can be retrieved.	To provide more flexibility for Issuers to configure the card.	Optional for Issuers	15.5 A.2.9 B.11
Added a description about the padding requirement of the data element in the transaction log,	Enhance the Specifications.	N/A	16.2
Deleted the following description of CVM List (tag '8E'): "UnionPay defines: 100000 = Present Cardholder ID".	Remove the content which is not supported in the UICS 2017.	N/A	A.1

Basic Specifications – Part V Contactless Integrated Circuit Card Payment Specification

Description	Reason	Impact	Section
Added the requirement that the terminal shall not set any bits of TVR during transaction processing.	Enhance the Specifications.	N/A	4.2.1

Description	Reason	Impact	Section
Added PAR in GPO response data element list; added PAR in recommended signature data list.	Add PAR usage in qUICS.	Optional for Issuers and Acquirers	8.4.2.2 8.4.2.3
Deleted “low-value or CTTA check” related content, as the UICS 2017 does not support this check anymore.	Remove the content which is not supported in the UICS 2017.	N/A	8.4.3.8
Updated the access requirement of the tags ‘9F68’, ‘9F6C’, ‘9F79’, ‘9F77’, ‘9F6D’, ‘9F78’ that those data value can be updated via PUT DATA command.	Enhance the Specifications.	Optional for Issuers	A.6
Revised the online-only parameter configuration requirement.	Enhance the Specifications.	N/A	F.2

Basic Specifications –Terms and Definitions

Change	Reason	Impact	Section
Added definitions of CDCVM, ODA for online authorization and PAR.	Enhance the Specifications.	N/A	3.1

4.2.4 Test and Certification Requirements

The test plan needs to be updated in the case of non-compliance.